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Comparative study of different methods for bathymetric mapping based on multi-source remote sensing data

Tianqi Lu[1,2], Luyi Wang[1,2]✉, Caiying Ji[1,2], Xining Liu[1,2], Xuanyu Bai[1,2] & Changgao Shao[1,2]

Satellite-derived bathymetry (SDB) technology for mapping shallow water around islands and reefs plays a crucial role in the management of coastal zone resources. The accuracy of SDB depends strongly on the selection of satellite remote sensing data and bathymetric models. This study evaluated the performance of five SDB inversion methods, including the log-transformed linear model, log-transformed band-ratio model, k-nearest neighbor regression, random forest, and support vector regression, using Landsat-8, Sentinel-2, and WorldView-2 imagery together with measured bathymetric data from Lingyang Reef. We compared and analyzed the bathymetric results of remote sensing data with different spatial resolutions in empirical models and machine learning models. The results show that the bathymetric results derived from different data across all models were able to reflect the spatial variation trends of water depth. The machine learning models achieved higher overall accuracy than the empirical models, with the random forest performing best across different data conditions. Sentinel-2 data exhibited greater stability in empirical model fitting, while WorldView-2 imagery achieved higher bathymetric accuracy when used with machine learning models. This study provides an empirical evidence for the appropriate selection of multispectral imagery with varying spatial resolutions and corresponding shallow water bathymetry models, contributing to enhanced accuracy and broader applicability of SDB.

Keywords Satellite-derived bathymetry, Different spatial resolution, Remote sensing imagery, Empirical model, Machine learning model

Shallow water bathymetric mapping of islands and reefs is crucial for navigational safety, coastal engineering, disaster warning, and coral reef ecological assessment and protection[1–4]. Compared with traditional bathymetry methods, such as shipborne sonar and airborne LiDAR, satellite-derived bathymetry (SDB) offers wide coverage, frequent revisit capability and low cost[5,6]. SDB is particularly valuable for offshore and remote waters where field surveys are constrained by challenging environmental conditions.

Satellite remote sensing technology has been used to bathymetric mapping since the 1970s[7]. In recent decades, multispectral satellite data with spatial resolutions ranging from several meters to hundreds of meters have been widely used in SDB studies. Satellite data with coarse spatial resolution, such as MODIS[8,9], SeaWiFS[10] and VIIRS[11], are less commonly used in SDB because of low spatial resolution (> 100 m). In contrast, moderate spatial resolution (≥ 10 m) satellite data, such as Sentinel-2[12–17] and the Landsat series[18–21], have become the most widely used data sources for SDB because of their acceptable spatial resolution and free access. With the innovation of remote sensing technology, high spatial resolution (< 10 m) satellite data (e.g. WorldView series, GaoFen series, Dove, GeoEye-1, etc.) have further improved the spatial detail of bathymetric mapping[6,22–28].

Satellite data with different spatial resolutions have been applied in SDB. However, the influence of image spatial resolution on bathymetric accuracy remains unclear. Although high spatial resolution imagery can reduce mixed pixel effects and provide more detailed bathymetry information, SDB performance is also strongly affected by the spectral characteristics and signal to noise ratios of the satellite data[27]. Therefore, higher spatial resolution does not necessarily guarantee better bathymetric accuracy under all conditions. A systematic comparison of multi-source remote sensing data with different spatial resolutions is still needed in SDB research.

[1]Sanya Institute of South China Sea Geology, Guangzhou Marine Geological Survey, China Geological Survey, Sanya, China. [2]Key Laboratory of Marine Mineral Resources, Ministry of Natural Resources, Guangzhou Marine Geological Survey, China Geological Survey, Guangzhou, China. ✉email: sanyalandsat@126.com

Bathymetric inversion performance is jointly influenced by both remote sensing data and inversion models. Over the past several decades, a variety of SDB methods has been developed to improve the accuracy and reliability of bathymetric results. Existing SDB approaches can generally be divided into two major categories: empirical models and analytical models. The log-transformed linear model (LLM)[29,30] and the log-transformed ratio model (LRM)[31] are the most representative empirical algorithms, which establish relationships between spectral reflectance and water depth through log-linear or log-ratio linear regression equations[3,12]. These empirical models are computationally efficient and have high accuracy in bathymetry compared with the analytical models, and have been widely used in various bathymetric scenes[32–34]. However, the relationship between the electromagnetic radiance received by satellite sensors and water depth is highly complex and affected by multiple factors, including water optical properties, bottom reflectance, and atmospheric effects. Therefore, modeling it with a linear formula may not be efficient[26,34,35].

The development of machine learning technology has provided a new approach for SDB. Machine learning algorithms, such as Random Forest (RF)[36–38], Support Vector Machine (SVM)[39–41] and different architectures of neural networks[13,42,43], have demonstrated strong capability in modeling complex nonlinear relationships between spectral reflectance and water depth[34]. Most previous studies have primarily focused on a single satellite data or a limited number of inversion methods. Systematic comparisons among different bathymetric models using multi-source remote sensing data with different spatial resolutions remain limited. Meanwhile, the variation of model performance across different water depth intervals has not been sufficiently investigated. Therefore, a comprehensive evaluation of different bathymetric models using multi-source remote sensing data is still needed to better understand the applicability and limitations of different SDB approaches.

In this study, multispectral satellite imagery with different spatial resolutions was used to map shallow water bathymetry around Lingyang Reef of Xisha Islands. Five bathymetric models were employed to evaluate the inversion accuracy using satellite images of different spatial resolutions, while in-situ depth measurements were used to assess the inversion performance across various depth ranges. This study aims to compare the bathymetric performance of multi-source remote sensing imagery with different spatial resolutions, evaluate the applicability of empirical and machine learning models for shallow water bathymetry, and analyze the depth-dependent variation characteristics of different SDB models.

Materials

Study area

The study area is located in the waters surrounding Lingyang Reef (111°35′22″E, 16°27′52″N), which is situated in the western part of the Xisha Islands in the South China Sea (Fig. 1). Lingyang Reef has itself the shape of an

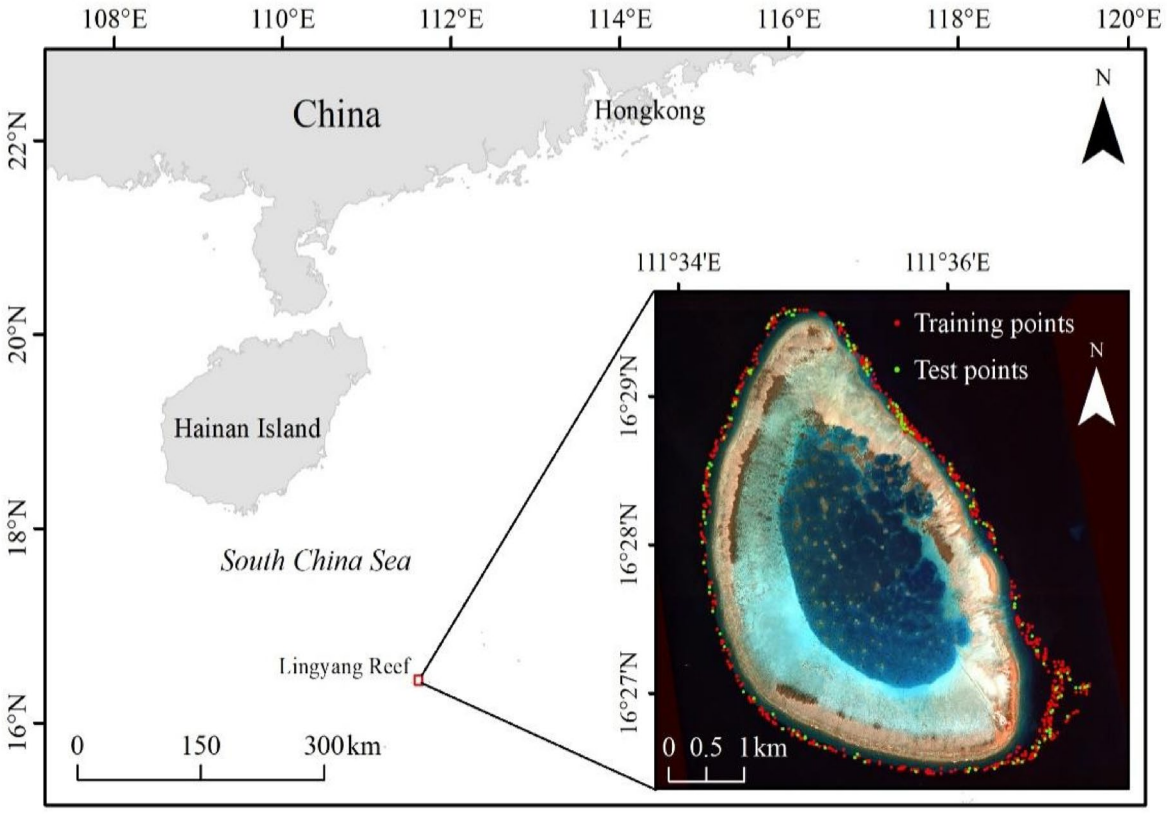


Fig. 1. Location of Lingyang Reef. Maps were drawn by authors, using ArcGIS 10.8 (Environmental Systems Research Institute, USA. https://www.esri.com).

isolated, small atoll with a central lagoon. The reef is approximately triangular, 6 km long from north to south, 4 km wide from west to east, is surrounded by coral reef.

Lingyang Reef is situated within a tropical monsoon climate and is almost unaffected by human activities, and has transparent waters. The water environmental condition is suitable for the growth of coral reefs and is an ideal area for satellite bathymetric research[6].

Satellite multispectral imagery

Satellite multispectral image data of three spatial resolutions were employed in this study, including Landsat-8 (L8), Sentinel-2 (S2) and WorldView-2 (WV2).

The L8 and S2 represent two of the most widely utilized remote sensing platforms in contemporary remote sensing science. L8 and S2 were launched in February 2013 and June 2015, respectively. Currently, they provide the best freely available multispectral datasets. The L8 Level-1T (L1T) image[44] and S2 Level-1 C (L1C)[45] image, both standard geometrically corrected products, were used in this study. L8 L1T image can be obtained from the website of United States Geological Survey (https://earthexplorer.usgs.gov), whereas S2 L1C image can be acquired from and Copernicus Browser (https://browser.dataspace.copernicus.eu). The spatial resolutions of the L8 and S2 satellites in the red, green, blue and near-infrared bands are 30 m and 10 m, respectively.

WorldView-2, one of the most superior commercial remote sensing satellite in the current market context, was launched in October 2009. It provides an eight-band multispectral imagery with a resolution of up to 1.84 m[46]. This study used four visible bands of WV2 that are consistent with the wavelengths of L8 and S2. Further satellite image technical details are provided in Table 1.

Measured water depth data

The measured water depth data were derived from the Guangzhou Marine Geological Survey, China Geological Survey, which were collected by the multi-beam sounding system aboard a geological survey vessel in September 2019. Tidal correction was applied by normalizing all measured depths to mean sea level, thereby reducing the influence of tidal fluctuations and ensuring consistency between the satellite imagery and in situ bathymetric measurements. The acquisition dates of the satellite imagery and the measured data are nearly coincident. Therefore, the measured data can be reliably applied in this study.

Methods

Workflow

This study employs different spatial resolutions image data, L8, S2 and WV2, for bathymetric mapping and accuracy evaluation, and the specific operation process is shown in Fig. 2. First, the L8 and WV2 images were geo-corrected with S2 L1C image after radiometric calibration, atmospheric correction. The blue, green, red and near-infrared bands were extracted to form the new image data. Second, the normalized differences water index (NDWI)[47,48] value was calculated to extract the waterbody range, followed by sun glint correction if necessary. The water-leaving reflectance (R_w) was retrieved after the processes above. Finally, bathymetric mapping of the study area was completed based on five different models using R_w data and 80% of the measured bathymetry data, and an accuracy evaluation was performed using 20% of the measured bathymetry data.

Satellite image data preprocessing

Radiometric calibration and atmospheric correction for L8 and WV2 were performed in ENVI (Environment for Visualizing Images) 5.6 software, converting the digital number values of imagery to the remote sensing reflectance just above the water surface (R_{rs}). The fast line-of-sight atmospheric analysis of spectral hypercubes (FLAASH) atmospheric correction method was used to remove the influences of the atmosphere on L8 and WV2 images in this study. The atmospheric correction of S2 L1C image was carried out using Sen2Cor version 2.10 software obtained from the European Space Agency, resulting in the S2 Level-2 A image, which is known as R_{rs} in this study.

The sun glint phenomenon is the result of Fresnel reflection of sunlight on a rough sea surface caused by wind and waves. It appears as a white flare in optical remote sensing images[34]. Each of the images used in this study avoided the effects of sun glint calculated by the Cox–Munk model using the sun-satellite geometric relationship[49,50], and it was not necessary to perform sun glint correction. Therefore, the R_w can be approximately equal to R_{rs} in this study.

Satellite image	Landsat-8 L1T	Sentinel-2 L1C	WorldView-2
Images date	08/10/2019 T02:54:13	07/24/2019 T02:55:51	07/09/2019 T03:04:40
Wavelength (nm)	482.6 (Blue) 561.3 (Green) 654.6 (Red) 864.6 (Near-infrared)	496.6 (Blue) 560.0 (Green) 664.5 (Red) 835.1 (Near-infrared)	480.0 (Blue) 545.0 (Green) 660.0 (Red) 832.5 (Near-infrared)
Spatial resolution (m)	30	10	2
Tide level (m)	1.1	1.1	1.2

Table 1. Specifics of the data used in this study.

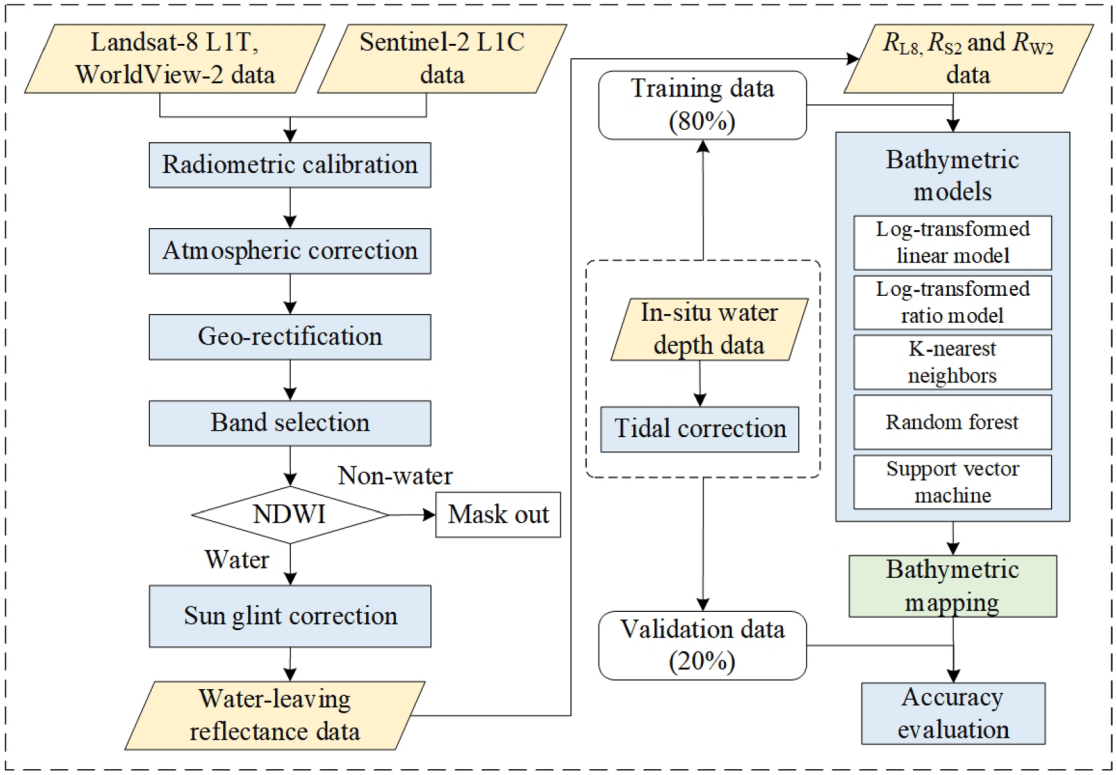


Fig. 2. Flow diagram of the bathymetric mapping. R_{L8}, R_{S2} and R_{W2} are the water-leaving reflectance obtained of Landsat-8, Sentinel-2 and WorldView-2, respectively.

Measured bathymetric data processing

Based on the remote sensing bathymetric capability, a total of 3362 measured points ranging from 0 to 30 m were selected. We used the subset features tool in ArcMap 10.8 to reduce the density of the measured points to minimize the occurrence of multiple measured points in one pixel, and 718 points retained. The measured data were randomly divided into training (80%) and test (20%) samples according to the preset ratio (Fig. 1). The training sample contained 574 points, while the test sample contained 144 points. The training sample was employed to develop the bathymetric models, and the test sample was used to evaluate the accuracy of the bathymetric results.

Bathymetric models

We used two empirical bathymetric models (log-transformed linear model and log-transformed ratio model) and three machine learning bathymetric models (k-nearest neighbors, random forest and support vector machine) to derive bathymetric maps.

Empirical models

Log-transformed linear model (LLM) The LLM assumes that the ratio of the bottom reflectance in the two image bands is consistent[29,30,51]. The effects of atmospheric scattering and water surface reflectance are eliminated by deducting the deep water reflectance as shown in Eq. (1).

$$Z = a_0 + \sum {}_{i=1}^{n} a_i \ln[R_w(\lambda_i) - R_\infty(\lambda_i)] \quad (1)$$

where Z is the water depth derived from the image, n is the number of bands, a_0 and a_i are the regression parameters to be solved, $R_w(\lambda_i)$ is the R_w for the band i, and $R_\infty(\lambda_i)$ is the deep water reflectance for the band i.

Log-transformed ratio model (LRM) The LRM estimates the depth value by comparing the two bands according to the principle that the radiation attenuation rate differs in different bands[31]. Its general form is given by Eq. (2).

$$Z = m_1 \frac{\ln(R_w(\lambda_i))}{\ln(R_w(\lambda_j))} + m_0 \quad (2)$$

where m_1 and m_0 represent the regression coefficients to be solved, $R_w(\lambda_i)$ and $R_w(\lambda_j)$ are the R_w for two different spectral bands.

Machine learning models

K-nearest neighbors (KNN) KNN is a concise and efficient nonparametric machine learning algorithm that utilizes the weighted average of the K nearest points for prediction[52]. It predicts the depth of each pixel by identifying the K most similar training points (based on Euclidean distance), and then averaging their corresponding target values[53,54]. In this study we used the $K = 5$. The instance-based learning paradigm of this algorithm enables flexible adaptation to complex, nonlinear relationships.

Random forest (RF) RF is a common ensemble learning method, which improves the accuracy of model prediction by constructing multiple decision trees and integrating the prediction results[55,56]. The RF method can effectively reflect the nonlinear relationships between variables in the water depth inversion, and has shown excellent performance in SDB[57]. In this study, the R_w in the red, green, blue and near-infrared bands selected as the model input, and a value of 300 was set to the number of trees.

Support vector machine (SVM) SVM is a supervised machine learning algorithm widely utilized in SDB for its ability to discern the nonlinear relationship between reflectance and water depth[57]. The algorithm maps the input data to a high-dimensional feature space through a nonlinear kernel function[58], establishing the optimal linear relationship between high-dimensional feature and water depth. The radial basis function (RBF) was used as kernel function for high-dimensional mapping in this study because of its superior generalization properties[59].

Accuracy evaluation

The accuracy of the SDB models was evaluated by the coefficient of determination (R^2), mean absolute error (MAE), root mean square error (RMSE) and mean absolute percentage error (MAPE), calculated as follows[60–63]:

$$R^2 = \frac{\left(\sum_{i=1}^{n}\left(D_i - \overline{D}_i\right)\left(Z_i - \overline{Z}_i\right)\right)^2}{\sum_{i=1}^{n}\left(D_i - \overline{D}_i\right)^2 \sum_{i=1}^{n}\left(Z_i - \overline{Z}_i\right)^2} \tag{3}$$

$$MAE = \frac{\sum_{i=1}^{n}|D_i - Z_i|}{n} \tag{4}$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n}(D_i - Z_i)^2}{n}} \tag{5}$$

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{D_i - Z_i}{D_i}\right| \times 100\% \tag{6}$$

where n is the number of test points; D_i is the measured water depth; Z_i is the water depth predicted by the model; $\overline{D}_i$ and $\overline{Z}_i$ are the average of the measured and predicted water depths, respectively.

Results

Empirical model establishment

The measured water depth values of training sample were taken as a dependent variable and the R_w values of the corresponding bands or band combinations were taken as independent variables, and the empirical models were developed according to the form of the LLM and LRM. The R^2 values of each model were calculated (Fig. 3) to evaluate the linear relationship between bands or band combinations and measured depths.

Figure 3 shows that there are significant differences in the relationships between water depth and input variables. From a longitudinal perspective, the LLMs exhibit relatively low R^2 when using a single spectral band as input, particularly in the red and near-infrared bands. This is because clear seawater strongly absorbs light at red and near-infrared wavelengths, resulting in very weak reflectance in those bands, and thus insufficient support for model fitting. The LLMs performance significantly improves when two or more band combinations are inputs except for the combination of the red and the near-infrared bands. Notably, the LLMs with four bands combination exhibit the highest fitting degree across all satellite images, and the R^2 values for L8, S2, and WV2 are 0.773, 0.840, and 0.821, respectively. Contrastingly, the LRMs for three satellite images perform poorly and unstable overall. The highest R^2 values are obtained from the combination of the green band with either the red or near-infrared band. Even so, they are still lower than those obtained when using only the green band as the input for the LLM.

A horizontal comparison reveals that S2 almost consistently yields the highest R^2 across various input schemes, including single bands, multiple bands, and band ratios. For instance, the LLM with four bands combination results in an R^2 of 0.839 for S2, which is the highest among all models. WV2 and L8 follow with 0.821 and 0.773, respectively. For LLMs using multi-band inputs, the performance of WV2 is generally better than that of L8. On the contrary, the performance of L8 is superior to that of WV2 in the LRMs. Across the 21 empirical models, eight for S2 exceed $R^2 = 0.800$ (and an additional four fall within 0.010 of 0.800). WV2 yields six models with $R^2 > 0.800$, and none of the L8 models surpass that threshold. Overall, empirical models constructed from S2 image show a distinct performance advantage.

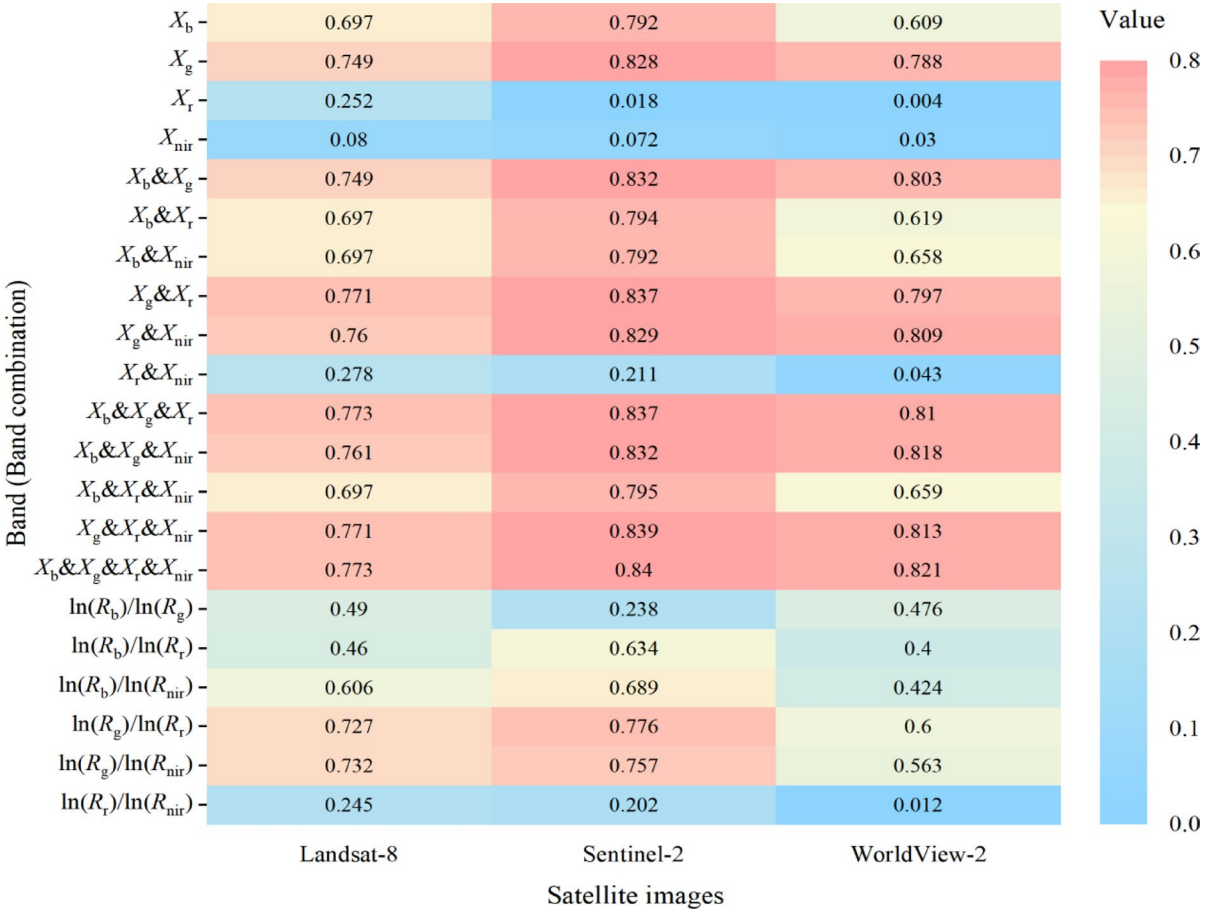


Fig. 3. Coefficient of determination of empirical models with different bands or band combinations. X_i is the natural logarithm of the difference between the water-leaving reflectance and the deep water reflectance for band i, R_i represents the water-leaving reflectance of band i.

Depth inversion results

The training of various SDB models and the subsequent depth inversion were carried out using the training samples together with the corresponding multi-band reflectance values. The models exhibiting the highest R^2 within the LLMs and LRMs were selected for bathymetry. Specifically, for the three satellite images, the optimal LLMs were those constructed using four bands (Fig. 4a–c). The LRM with the highest R^2 for L8 was developed using the ratio of the green to near-infrared bands as input (Fig. 4d), whereas for S2 and WV2, the LRMs with the highest R^2 were established based on the ratio of the green to red bands (Fig. 4e, f).

Figure 5 shows the bathymetric results derived from three images using five SDB models. The water depth shown in each bathymetric map gradually increases from the reef towards the outside, which is consistent with the actual situation. Overall, the trends of water depth variation in each map exhibit a strong spatial consistency, although there are obvious differences in the numerical values.

To assess the performance of bathymetry estimation of the five models, the test samples were used to calculate the MAE, RMSE and MAPE of the various models (Table 2). For all SDB models, the bathymetry accuracy from the WV2 image is significantly superior to that of the L8 and S2 images, and the result with the highest precision is obtained by using the RF model on the WV2 image. According to MAE, RMSE and MAPE, the order of the overall bathymetry accuracy of the five SDB models using the L8 image is RF > KNN > LLM > LRM > SVM, the order for the S2 image is KNN > RF > LLM > SVM > LRM, and the order for the WV2 image is RF > KNN > LLM > SVM > LRM (Fig. 6). Except for SVM, the performance of machine learning models (RF and KNN) is superior to that of empirical models (LLM and LRM). This phenomenon is mainly because the machine learning models possess a strong ability for nonlinear fitting, and can to a certain extent adapt to the diverse radiation attenuation caused by water column effect and seafloor heterogeneity.

Discussion

According to the Lambert-Beer law, light decays exponentially with the increase in water depth as it travels through the water column[64,65]. In this scenario, the shallower the water depth, the higher the accuracy of remote sensing bathymetry; conversely, as depth increases the bathymetric accuracy declines sharply. The bathymetric results were segmented in 5 m intervals to explore the bathymetric accuracy across different depth ranges, and the corresponding MAE, RMSE and MAPE were calculated.

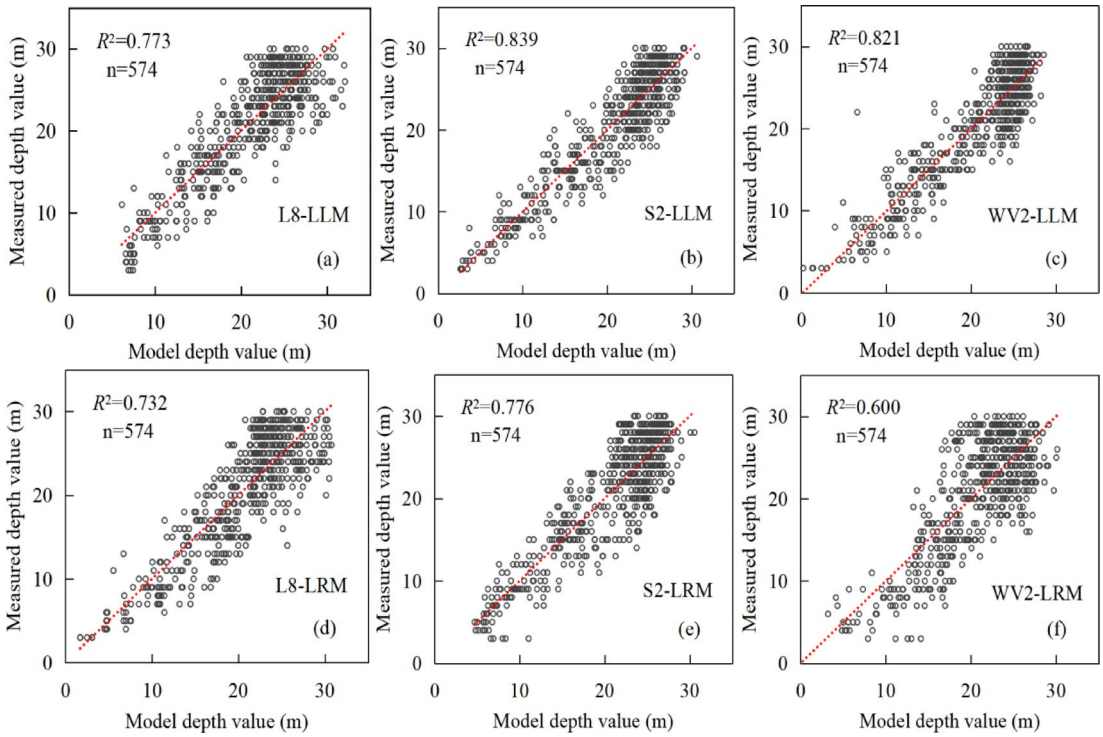


Fig. 4. Scatter plots of the model results and measured depth results. (**a–c**) are the results of the LLMs constructed using four bands from L8, S2 and WV2, respectively. (**d**) is the result of the LRM constructed from the green and near-infrared bands of L8. (**e,f**) are the results of the LRMs constructed from the green and red bands of S2 and WV2, respectively. The red dotted line is the 1:1 line.

Model accuracy analysis across varied depth ranges

Figure 7 shows the depth-dependent error variations of the five bathymetric models using different satellite data. Overall, the MAE and RMSE increase from shallow to intermediate water depths, and then gradually decrease or stabilize with further increases in depth. The increasing errors in the shallow to intermediate zones are primarily attributed to the exponential attenuation of bottom reflected signals through the water column. However, beyond a certain depth, the MAE and RMSE gradually stabilize and even exhibit a slight decline. This phenomenon deviates from the idealized Lambert-Beer law and has also been reported in previous SDB studies[1]. A plausible explanation is that the spectral response gradually loses sensitivity to depth variations in deeper waters due to the severe attenuation of bottom reflected signals. Contrastingly, the MAPE values of the L8 and S2 images first increase and then decrease with increasing depth, whereas the MAPE of the WV2 image shows an overall decreasing trend. This difference is mainly related to the characteristics of the relative error metric. In shallow waters, relatively small absolute depth deviations can produce large percentage errors because the reference depths are small. As water depth increases, the influence of the denominator effect gradually decreases, leading to a decline in MAPE values. In addition to water column attenuation, the increase in errors within the 5–10 m depth range may be related to substrate heterogeneity. Although the study area is dominated by coral reef environments, local variations in live coral, coral rubble, and sand patches can generate substantial differences in bottom reflectance. These spectral differences may be misinterpreted as depth variations, thereby increasing inversion uncertainty. The phenomenon is observed in all three data but is more pronounced in the lower resolution L8 and S2 imagery. In contrast, the smaller error increase observed for the WV2 data suggests that the higher spatial resolution can partially mitigate the influence of substrate heterogeneity by reducing mixed pixel effects and better resolving bottom characteristics.

The inversion accuracy of machine learning models is generally better than that of empirical models in different water depth intervals, indicating that machine learning algorithms possess stronger nonlinear fitting capability and robustness, enabling better modeling of the complex nonlinear relationship between water depth and multi-band reflectance[34,66].

The performance of individual machine learning models varies across different water depth ranges. The KNN and RF models perform the best overall, particularly achieving higher inversion accuracy in the 0–10 m and 25–30 m depth ranges. Contrastingly, the error of the SVM model exhibits larger fluctuations with increasing water depth, which may be related to the selection of the kernel function, parameter optimization, and its sensitivity to abnormal samples. In this study, the SVM did not demonstrate the advantages of the machine learning model. However, an interesting phenomenon is that the SVM model performed better than other models in the 20–25 m water depth range. This observation aligns well with Zhou et al.[67],. Among the empirical models, the LRM showed relatively larger error, probably because it relies on information from only two spectral bands, whereas

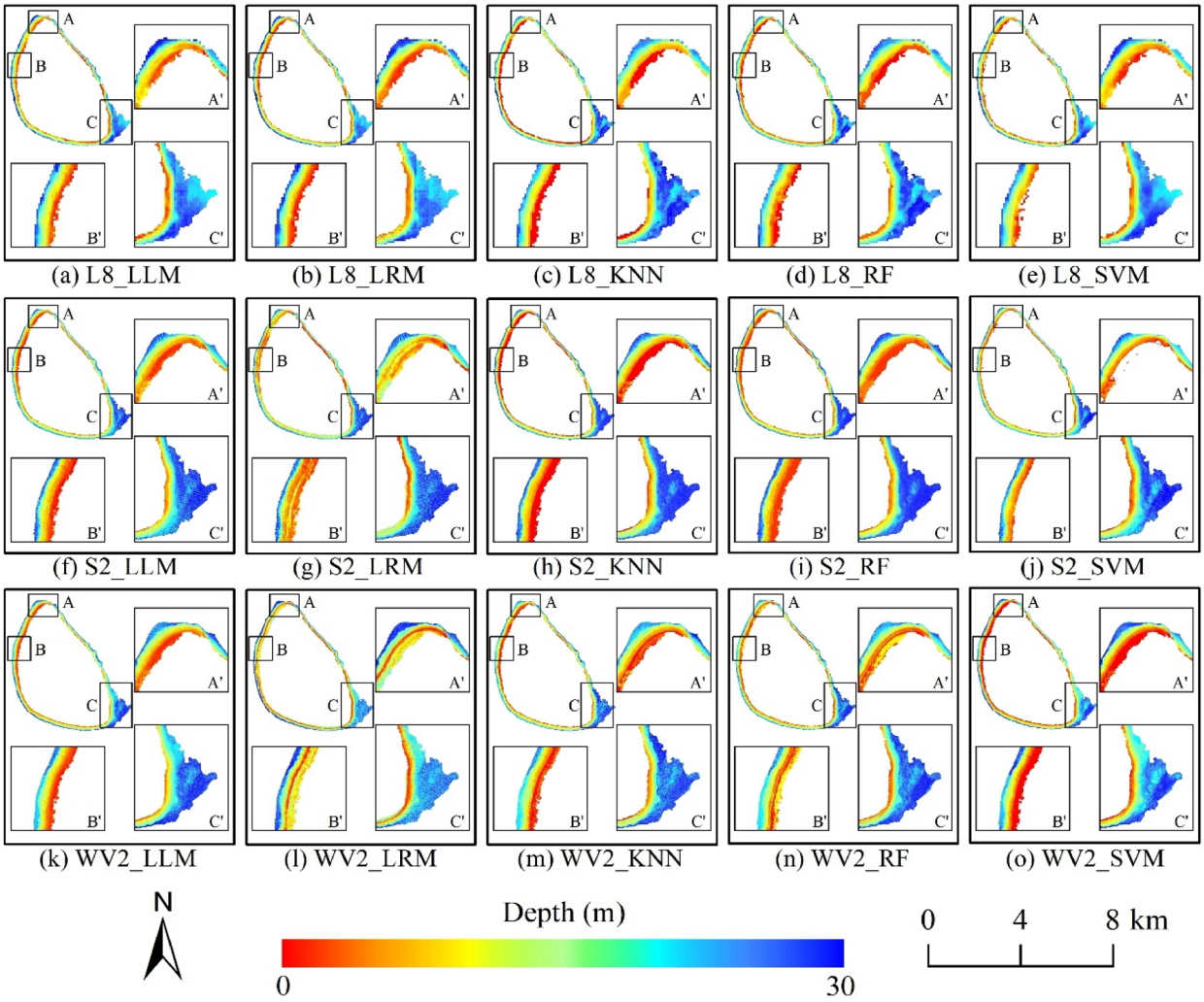


Fig. 5. Bathymetric maps derived from three images using five SDB models. A', B', and C' show enlarged views of regions A, B, and C, respectively. Maps were drawn by authors, using ArcGIS 10.8 (Environmental Systems Research Institute, USA. https://www.esri.com).

Accuracy indicator	Data	Model				
		LLM	LRM	KNN	RF	SVM
MAE (m)	L8	2.82	3.06	2.25	2.16	3.10
	S2	2.19	2.40	1.88	1.92	2.29
	WV2	2.00	3.22	1.87	1.86	2.51
RMSE (m)	L8	3.39	3.73	2.98	2.80	3.78
	S2	2.73	2.93	2.43	2.48	2.84
	WV2	2.47	4.05	2.35	2.32	3.05
MAPE (%)	L8	17.16	17.67	13.05	12.97	20.18
	S2	12.10	13.67	10.84	11.03	13.79
	WV2	12.32	20.45	10.39	9.99	15.91

Table 2. Accuracy of different bathymetric results.

the LLM incorporates a greater number of bands. The higher number of bands can provide more information about the water column and bottom conditions[12].

It can be observed that the relative errors of the KNN and RF models change relatively smoothly as the water depth range varies, especially on the WV2 data (Fig. 7i). This indicates that the KNN and RF models not only

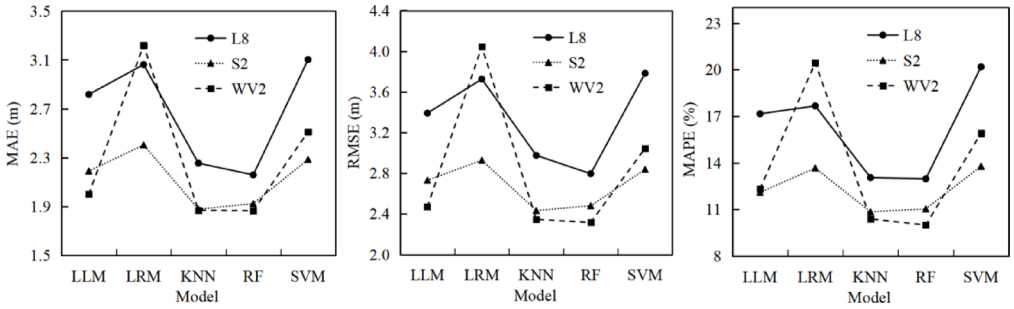


Fig. 6. Comparison of errors in different bathymetric results.

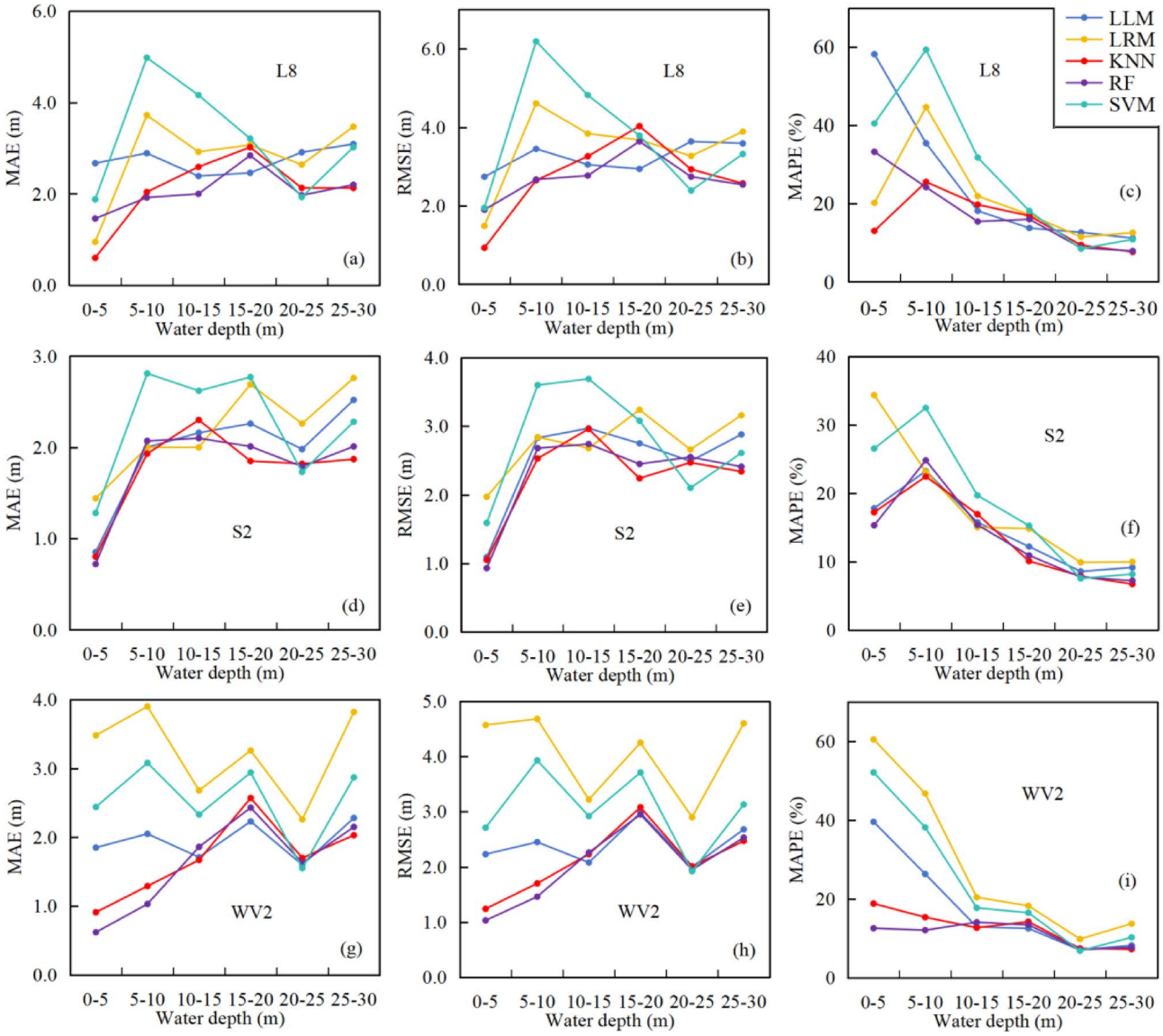


Fig. 7. Comparison of errors among different bathymetric models. (**a–c**) show error variations for the five models with L8 data, (**d–f**) for S2 data, and (**g–i**) for WV2 data.

perform well in controlling the absolute error, but also maintain good stability in controlling the relative error, demonstrating strong adaptability to varying spatial resolutions and spectral data.

Assessment of image performance across different water depth intervals

Figure 8 presents the bathymetric errors of the three remote sensing data in different models. Clear differences in bathymetric accuracy were observed among the three datasets, indicating that the spatial resolution and signal to noise ratios of the data sources are the key factors influencing depth inversion performance.

The performance of the three data varies greatly in the two empirical models. The absolute error of WV2 in LLM is generally lower and changes smoothly (Fig. 8a, b). In contrast, the bathymetric error of WV2 in LRM is higher, while the accuracy of S2 is significantly better than both L8 and WV2 (Fig. 8d, e). In general, the relative error of S2 in the empirical models changes more smoothly (Fig. 8c, f), indicating that relying solely on high spatial resolution is not sufficient to improve the accuracy of SDB in the empirical models. The higher signal to noise ratios of S2 may provide a more stable linear depth response, resulting in a smoother and more stable relative error change.

The advantage of high spatial resolution WV2 data is more pronounced in the machine learning models, with both absolute and relative errors of WV2 generally lower than those of S2 and L8 in the KNN and RF models. This indicates that high spatial resolution data can significantly enhance the fitting ability of nonlinear models and more effectively capture the complex change patterns between spectra and water depth. An interesting phenomenon is that, whether in empirical models or machine learning models, the accuracy of S2 within the water depth range of 15–20 m is the highest. This warrants further investigation into the reasons behind it.

Conclusion

This study demonstrates that, two empirical models (LLM and LRM) and three machine learning models (KNN, RF, and SVM) were established for shallow water depth inversion using multispectral satellite images with three different spatial resolutions (L8, S2, and WV2) and measured water depth data. The accuracy of each model was systematically compared and analyzed, and the following conclusions were obtained.

(1) Both empirical models and machine learning models are capable of inverting optically shallow water depth from multispectral images with different spatial resolutions. The inversion results can reflect the real water depth trends. The spatial resolution and signal to noise ratios have significant impacts on the accuracy of SDB.
(2) The LLM established with four bands has significantly better performance than other empirical models. S2 data demonstrates better fitting stability in the empirical models. Improving the spatial resolution alone has a limited effect on enhancing the bathymetric accuracy of the empirical models.
(3) The machine learning models, particularly KNN and RF, achieve substantially higher overall accuracy than empirical models. For both KNN and RF models, bathymetric accuracy generally improves with increasing image spatial resolution.
(4) The machine learning models, especially KNN and RF, demonstrate higher accuracy and stability across different water depth ranges. In these two models, the high spatial resolution WV2 imagery consistently yields superior and more stable bathymetric performance across varying depth intervals. Substrate heterogeneity is an important source of inversion uncertainty within the 5–10 m depth range, particularly for lower resolution imagery.

Based on the conclusions above, this study provides a practical reference for shallow water bathymetric inversion in optically clear waters using multispectral imagery and measured bathymetric data, particularly with respect to the selection and construction of models based on different spatial resolution satellite data. This will greatly facilitate the use of SDB technology for the management of shallow water areas. Future work will evaluate the proposed models in different marine environments and will investigate the influence of substrate heterogeneity through benthic classification.

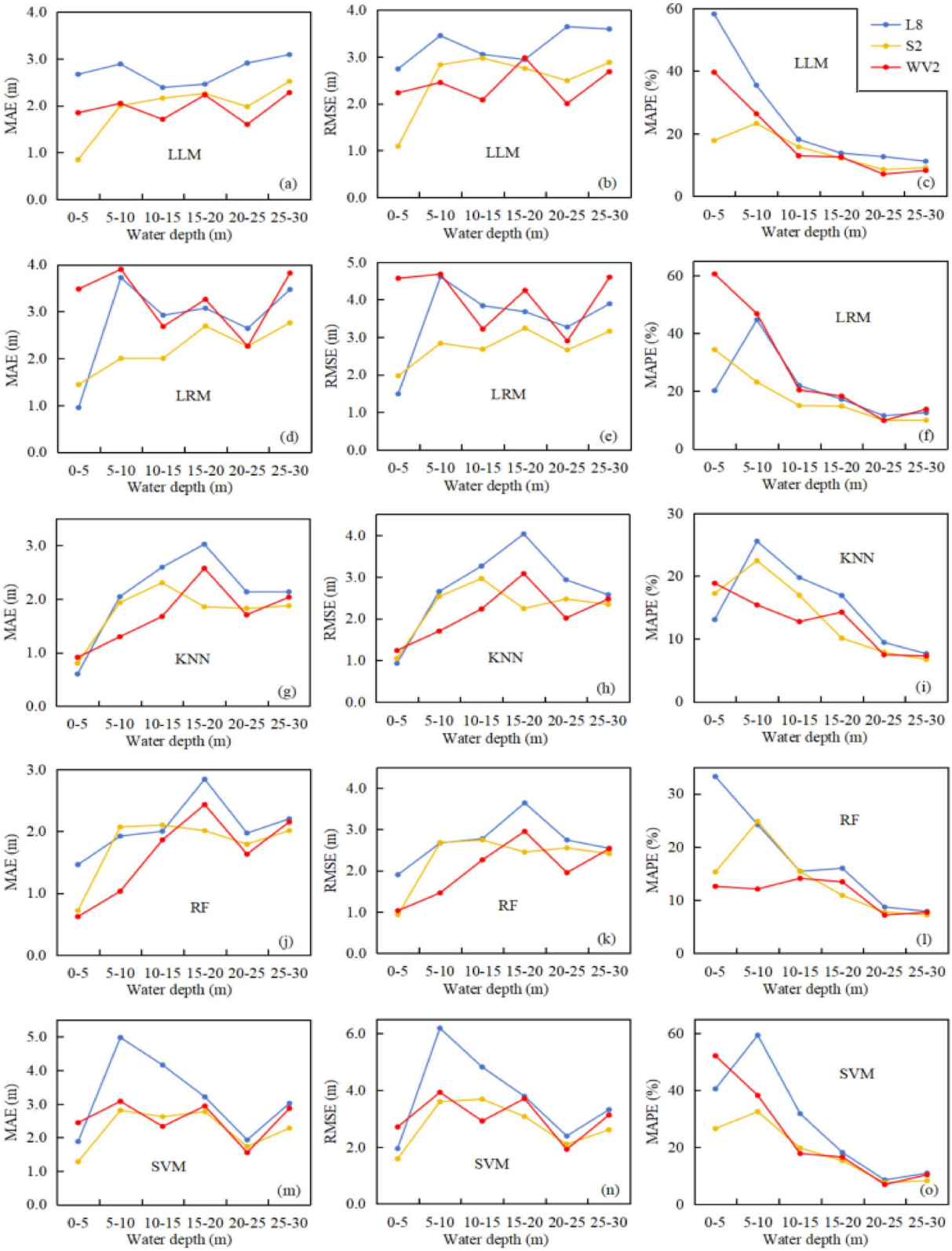


Fig. 8. Comparison of bathymetric errors among different satellite data. (**a–c**) show error variations of the LLM for the three data, (**d–f**) of the LRM model, (**g–i**) of the KNN model, (**j–l**) of the RF model, and (**m–o**) of the SVM model.

Data availability

The data that support the findings of this study are available from Maxar Technologies but restrictions apply to the availability of these data, which were used under license for the current study, and so are not publicly available. Data are however available from the authors upon reasonable request and with permission of Maxar Technologies. The remaining datasets used and analysed during the current study available from the corresponding author on reasonable request.

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Author contributions

T. Lu contributed to conceptualisation, methodology, data curation, data analysis, writing the original draft, reviewing and editing the manuscript. L. Wang contributed to conceptualisation, methodology, data validation, reviewing and editing the manuscript. C. Ji contributed to methodology, data validation, reviewing and editing the manuscript. X. Liu contributed to methodology, data curation, data validation, reviewing and editing the manuscript. X. Bai contributed to data curation, data validation, reviewing and editing the manuscript. C. Shao contributed to reviewing and editing the manuscript.

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Declarations

Competing interests
The authors declare no competing interests.

Additional information
Correspondence and requests for materials should be addressed to L.W.

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